

SAPIENZA
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Combinatorial Optimization

Volume 6 — Specialized algorithms

Chapters 29 of the module «Combinatorial Optimization»: specialized algorithms. Definitions, results with proofs and worked examples; exercises with complete solutions are in Exercises 6.

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Chapter 29

Knapsack: branch-and-bound on the critical item

Prerequisites

- From the module *Integer linear programming*, Chapter 7 «Branch-and-bound»: branching and bounding operations, the problem list, the pruning criteria, the lower bound LB and the upper bound UB. The general algorithm is applied here to the knapsack problem.
- From this module, the chapter «Knapsack formulations»: the models ILP_{KP} and LP_{KP}, the efficiency of an item, the split item s, the residual capacity rc and the bound of Dantzig.

Objectives

At the end of this chapter you will be able to:

- write the problem LP^k_{KP} of a node, and solve it with the formulas of Dantzig, without the simplex algorithm;
- branch on the variable of the split item, unselecting or selecting that item;
- execute the branch-and-bound algorithm for the KP by hand, with the FIFO or the LIFO rule;
- explain why the algorithm is simpler than the general one, and what stays the same.

29.1 The algorithm for the KP

Definition 29.1: branching node and $\mathrm{LP}^k_{\mathrm{KP}}$ problem

In the branch-and-bound algorithm for the $\mathrm{ILP}_{\mathrm{KP}}$ model, the branching constraints on a variable x_j, $j \in \mathscr{I}$, are $x_j \leq 0$ and $x_j \geq 1$. In a branching node k of the branch-and-bound tree, we have:

1. a subset of **unselected items** $\rightsquigarrow$ denoted $\mathscr{I}^k_{\mathrm{out}}$
2. a subset of **selected items** $\rightsquigarrow$ denoted $\mathscr{I}^k_{\mathrm{in}}$
3. a subset of **remaining items** $\rightsquigarrow$ denoted $\mathscr{I}^k := \mathscr{I} \setminus \left\{\mathscr{I}^k_{\mathrm{out}} \cup \mathscr{I}^k_{\mathrm{in}}\right\}$

The $\mathrm{LP}_{\mathrm{KP}}$ problem associated to node k, called the $\mathrm{LP}^k_{\mathrm{KP}}$ problem, reads as follows:

$$
\begin{aligned}
\max \quad & \sum_{j \in \mathscr{I}} p_j \, x_j \\
\text{subject to} \quad & \sum_{j \in \mathscr{I}} w_j \, x_j \leq c, \\
& x_j \geq 1, && \forall j \in \mathscr{I}^k_{\mathrm{in}}, \\
& x_j \leq 0, && \forall j \in \mathscr{I}^k_{\mathrm{out}}, \\
& x_j \leq 1, && \forall j \in \mathscr{I}, \\
& x_j \geq 0, && \forall j \in \mathscr{I}.
\end{aligned}
$$

- The branching constraints $x_j \leq 0$ and $x_j \geq 1$ have the effect of **unselecting** or **selecting** an item.

A $\mathrm{LP}^k_{\mathrm{KP}}$ problem of a branching node k can be equivalently re-written as:

$$
\begin{aligned}
& \underbrace{\sum_{j \in \mathscr{I}^k_{\mathrm{in}}} p_j}_{=p(k)} + \max \quad \sum_{j \in \mathscr{I}^k} p_j \, x_j && && \text{(29.1a)} \\
& \text{subject to} \quad \sum_{j \in \mathscr{I}^k} w_j \, x_j \leq \underbrace{c - \sum_{j \in \mathscr{I}^k_{\mathrm{in}}} w_j}_{=c(k)}, && && \text{(29.1b)} \\
& \qquad x_j \leq 1, && \forall j \in \mathscr{I}^k, && \text{(29.1c)} \\
& \qquad x_j \geq 0, && \forall j \in \mathscr{I}^k. && \text{(29.1d)}
\end{aligned}
$$

- A $\mathrm{LP}^k_{\mathrm{KP}}$ problem of a branching node k is **structurally equivalent** to the $\mathrm{LP}_{\mathrm{KP}}$ problem but:

1. the set of items is $\mathscr{I}^k$ instead of $\mathscr{I}$
2. the capacity is reduced to $c(k) := c - \sum_{j \in \mathscr{I}^k_{\text{in}}} w_j$
3. there is an objective-function constant $p(k) := \sum_{j \in \mathscr{I}^k_{\text{in}}} p_j$

29.1.1 Solving the LP^k_{KP} problem of a node

- The problem LP^k_{KP} is a knapsack LP problem on the items of $\mathscr{I}^k$, with capacity $c(k)$: it is solved by the formulas of Dantzig, without any simplex iteration. The items of $\mathscr{I}^k$ are kept in the order of non-increasing efficiency.
- If $c(k) < 0$ the problem is **infeasible**. If $\sum_{j \in \mathscr{I}^k} w_j \leq c(k)$ all the items of $\mathscr{I}^k$ fit, and the optimal solution is $\tilde{x}_j = 1$ for all $j \in \mathscr{I}^k$: it is integer.
- Otherwise the **split item** s of the node is the first item of $\mathscr{I}^k$ that does not fit entirely, and the **residual capacity** is the capacity left by the items that precede it:

$$\sum_{j \in \mathscr{I}^k, j<s} w_j \leq c(k) < \sum_{j \in \mathscr{I}^k, j \leq s} w_j, \qquad rc = c(k) - \sum_{j \in \mathscr{I}^k, j<s} w_j.$$

- The optimal solution is $\tilde{x}_j = 1$ for the items of $\mathscr{I}^k$ before s, $\tilde{x}_s = \frac{rc}{w_s}$ and $\tilde{x}_j = 0$ for the items after s, and the optimal value is

$$z(\text{LP}^k_{\text{KP}}) \;=\; p(k) + \sum_{j \in \mathscr{I}^k, j<s} p_j + p_s \frac{rc}{w_s}.$$

In every node at most one variable is fractional, the one of the split item, and the value of the node is the Dantzig bound of a smaller knapsack: the node is solved by a scan of the items.

29.1.2 Branching-variable selection strategy

- The optimal solution of an LP^k_{KP} problem given by the formulas of the previous subsection has at most one fractional variable, the one associated to the split item. **Branching can be performed on the variable associated to the split item**: this branching-variable selection strategy is called split-item strategy.
- The optimal solution of an LP_{KP} problem is not unique when some items have equal efficiency: the other optimal solutions can have more fractional variables, for instance $x = (\frac{1}{2}, \frac{1}{2})$ for $p = (2, 2)$, $w = (2, 2)$, $c = 2$. Here, as in the general algorithm, we always use the solution given by the formulas.

29.1.3 What changes with respect to the general algorithm

- The algorithm is the LP-based branch-and-bound algorithm of the module *Integer linear programming* (Chapter 7 «Branch-and-bound»), with its problem list, its pruning criteria and its optimality gap.
- **Bounding.** The problem of a node is solved by the formulas above. All the profits are integer, so the bound of a node is $\lfloor z(\mathrm{LP}^k_{\mathrm{KP}}) \rfloor$: the node is pruned by bound if $\lfloor z(\mathrm{LP}^k_{\mathrm{KP}}) \rfloor \leq \mathrm{LB}$, by infeasibility if $c(k) < 0$, by integrality if its solution is integer (then LB is updated when the value is larger).
- **Branching.** The branching variable is the one of the split item s: the children are $x_s \leq 0$ (down, item unselected) and $x_s \geq 1$ (up, item selected), and the node is not solved with a simplex tableau but described by the sets $\mathscr{I}^k_{\mathrm{in}}$, $\mathscr{I}^k_{\mathrm{out}}$ and $\mathscr{I}^k$.
- **Node selection.** The node of the problem list is extracted with the FIFO or the LIFO rule, and the down-branching node is solved before the up-branching node; the nodes are numbered in the order of resolution. The lower bound written in a node is the value at the moment of its resolution.

29.2 A worked example

Example 29.1: The $\mathrm{ILP}_{\mathrm{KP}}$ model

We consider a knapsack of capacity $c = 9$ and a set $\mathscr{I} = \{1, 2, 3, 4, 5, 6\}$ of 6 items ($n = 6$), with the following profits and weights:

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	$j=6$
p_j	6	5	8	9	6	7
w_j	2	3	6	7	5	9

We refer to this KP instance as Example 1. The $\mathrm{ILP}_{\mathrm{KP}}$ model for this KP instance is:

$$\begin{array}{rrrrrrrrrrrrrrr} \max & 6 & x_1 & +5 & x_2 & +8 & x_3 & +9 & x_4 & +6 & x_5 & +7 & x_6 & & \\ \text{subject to} & 2 & x_1 & +3 & x_2 & +6 & x_3 & +7 & x_4 & +5 & x_5 & +9 & x_6 & \leq & 9 \\ & & x_1, & & x_2, & & x_3, & & x_4, & & x_5, & & x_6 & \in & \{0,1\} \end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{KP}})$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 1, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$

The optimal value $z(\mathrm{ILP}_{\mathrm{KP}})$ is 15. The associated optimal solution of the KP is $\tilde{S} = \{1, 4\}$.

The items are sorted by non-increasing efficiency, since:

$$\frac{6}{2} \geq \frac{5}{3} \geq \frac{8}{6} \geq \frac{9}{7} \geq \frac{6}{5} \geq \frac{7}{9}$$

Example 29.2: Branch-and-bound algorithm for the $\mathrm{ILP}_{\mathrm{KP}}$ model

A branch-and-bound algorithm for $\mathrm{ILP}_{\mathrm{KP}}$ model and the KP instance of Example 1 using:

1. the split-item strategy as branching-variable selection strategy;
2. the FIFO rule as branching-node selection strategy;
3. solve first the down-branching node and then the up-branching node;
4. set the incumbent solution $\bar{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{KP}})$ to the following heuristic solution:

$$\bar{x}_1 = 1, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1, \quad \bar{x}_5 = 0, \quad \bar{x}_6 = 0$$

and set the lower bound LB accordingly.

- **Initialization**

  For the $\mathrm{LP}^1_{\mathrm{KP}}$ problem, the split item is $s = 3$ and the residual capacity is $rc = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^1_{\mathrm{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{2}{3}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$

  the optimal value is:

$$z(\,\mathrm{LP}^1_{\mathrm{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 8 \cdot \frac{2}{3} = \frac{49}{3}$$

  The optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^1_{\mathrm{KP}}\,)$ is **fractional** and the $\mathrm{LP}^1_{\mathrm{KP}}$ problem is **feasible**, so the $\mathrm{LP}^1_{\mathrm{KP}}$ problem is inserted in the problem list L and the lower bound LB is set to 15 ($\mathrm{LB}(\bar{\boldsymbol{x}}) = 15$).

  Due to the current lower bound LB and upper bound UB, we have:

$$15 \quad \leq \quad z(\mathrm{ILP}_{\mathrm{KP}}) \quad \leq \quad \underbrace{\left\lfloor \frac{49}{3} \right\rfloor}_{=16}$$

- **Iteration 1**

  The $\mathrm{LP}^1_{\mathrm{KP}}$ problem is extracted and deleted from the problem list L.

  The branching variable is x_3. The $\mathrm{LP}^2_{\mathrm{KP}}$ and $\mathrm{LP}^3_{\mathrm{KP}}$ problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

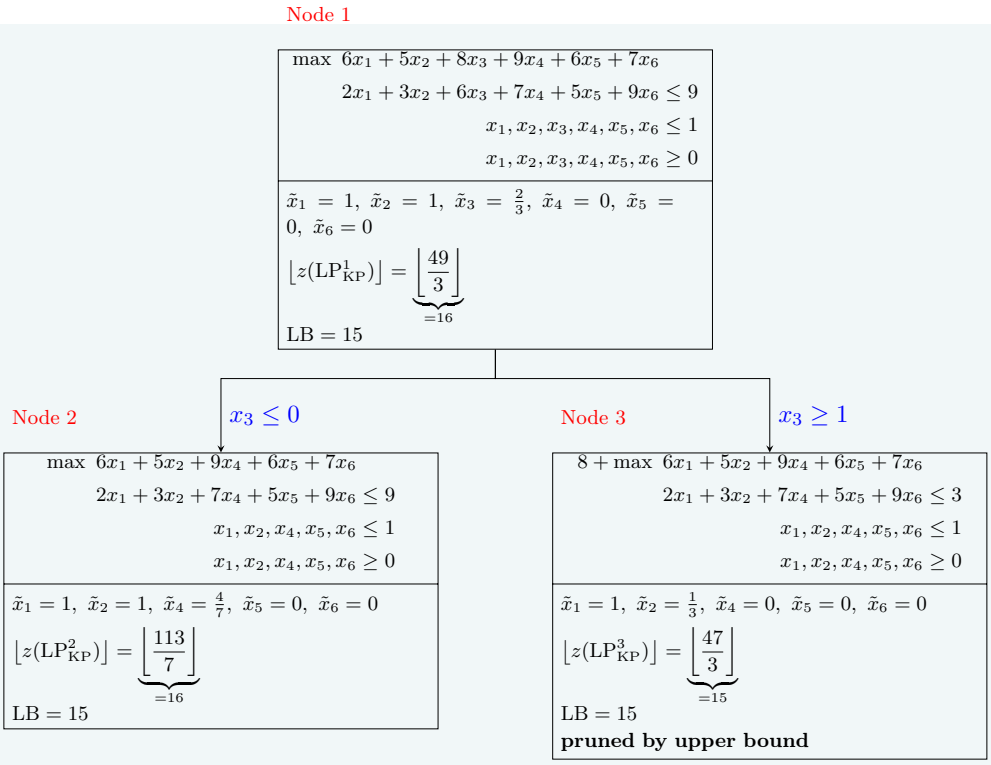
1. Solution of the $\mathrm{LP}^2_{\mathrm{KP}}$ problem

   We have
   $$\mathscr{I}^2 = \{1,2,4,5,6\}, \quad \mathscr{I}^2_{\mathrm{out}} = \{3\}, \quad \mathscr{I}^2_{\mathrm{in}} = \emptyset$$
   The capacity is $c(2) = 9$ and the objective-function constant is $p(2) = 0$. The split item is $s = 4$ and the residual capacity is $rc = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^2_{\mathrm{KP}}\,)$ is:
   $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_4 = \frac{4}{7}, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$
   and
   $$z(\,\mathrm{LP}^2_{\mathrm{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 9 \cdot \frac{4}{7} = \frac{113}{7}$$
   The $\mathrm{LP}^2_{\mathrm{KP}}$ problem is **feasible** and we have:
   $$\lfloor z(\mathrm{LP}^2_{\mathrm{KP}}) \rfloor > \mathrm{LB} \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{113}{7} \right\rfloor}_{=16} > 15$$
   The optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^2_{\mathrm{KP}}\,)$ is **fractional**, accordingly, the $\mathrm{LP}^2_{\mathrm{KP}}$ problem is **inserted in the problem list** L.

2. Solution of the $\mathrm{LP}^3_{\mathrm{KP}}$ problem

   We have
   $$\mathscr{I}^3 = \{1,2,4,5,6\}, \quad \mathscr{I}^3_{\mathrm{out}} = \emptyset, \quad \mathscr{I}^3_{\mathrm{in}} = \{3\}$$
   The capacity is $c(3) = 3$ and the objective-function constant is $p(3) = 8$. The split item is $s = 2$ and the residual capacity is $rc = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^3_{\mathrm{KP}}\,)$ is:
   $$\tilde{x}_1 = 1, \ \tilde{x}_2 = \frac{1}{3}, \ \tilde{x}_4 = 0, \ \tilde{x}_5 = 0, \ \tilde{x}_6 = 0 \quad \text{and} \quad z(\,\mathrm{LP}^3_{\mathrm{KP}}\,) = 8{+}6{\cdot}1{+}5{\cdot}\frac{1}{3} = \frac{47}{3}$$
   The $\mathrm{LP}^3_{\mathrm{KP}}$ problem is **feasible** and we have:
   $$\lfloor z(\mathrm{LP}^3_{\mathrm{KP}}) \rfloor \leq \mathrm{LB} \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{47}{3} \right\rfloor}_{=15} \leq 15$$
   accordingly, the branching node is pruned by upper bound.

At the end of the first iteration, Node 2 is inserted in the problem list L, while Node 3 is pruned by upper bound. The corresponding branch-and-bound tree is:



Due to the current lower bound LB and upper bound UB, we have:

$$15 \quad \le \quad z(\mathrm{ILP}_{\mathrm{KP}}) \quad \le \quad \underbrace{\left\lfloor \frac{113}{7} \right\rfloor}_{=16}$$

- **Iteration 2**

The $\mathrm{LP}^2_{\mathrm{KP}}$ problem is extracted and deleted from the problem list L.

The branching variable is x_4. The $\mathrm{LP}^4_{\mathrm{KP}}$ and $\mathrm{LP}^5_{\mathrm{KP}}$ problems are created by unselecting $(x_4 \le 0)$ and selecting $(x_4 \ge 1)$ item 4.

1. Solution of the $\mathrm{LP}^4_{\mathrm{KP}}$ problem

   We have

   $$\mathscr{I}^4 = \{1,2,5,6\}, \quad \mathscr{I}^4_{\mathrm{out}} = \{3,4\}, \quad \mathscr{I}^4_{\mathrm{in}} = \emptyset$$

   The capacity is $c(4) = 9$ and the objective-function constant is $p(4) = 0$. The split item is $s = 5$ and the residual capacity is $rc = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^4_{\mathrm{KP}}\,)$ is:

   $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_5 = \frac{4}{5}, \quad \tilde{x}_6 = 0 \quad \text{and} \quad z(\,\mathrm{LP}^4_{\mathrm{KP}}\,) = 6\cdot 1 + 5\cdot 1 + 6\cdot\frac{4}{5} = \frac{79}{5}$$

The $\mathrm{LP}^4_{\mathrm{KP}}$ problem is **feasible** and we have:

$$\lfloor z(\mathrm{LP}^4_{\mathrm{KP}}) \rfloor \leq \mathrm{LB} \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{79}{5} \right\rfloor}_{=15} \leq 15$$

accordingly, the branching node is pruned by upper bound.

2. Solution of the $\mathrm{LP}^5_{\mathrm{KP}}$ problem

We have

$$\mathscr{I}^5 = \{1, 2, 5, 6\}, \quad \mathscr{I}^5_{\mathrm{out}} = \{3\}, \quad \mathscr{I}^5_{\mathrm{in}} = \{4\}$$

The capacity is $c(5) = 2$ and the objective-function constant is $p(5) = 9$. The split item is $s = 2$ and the residual capacity is $rc = 0$. An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}^5_{\mathrm{KP}}\,)$ is:

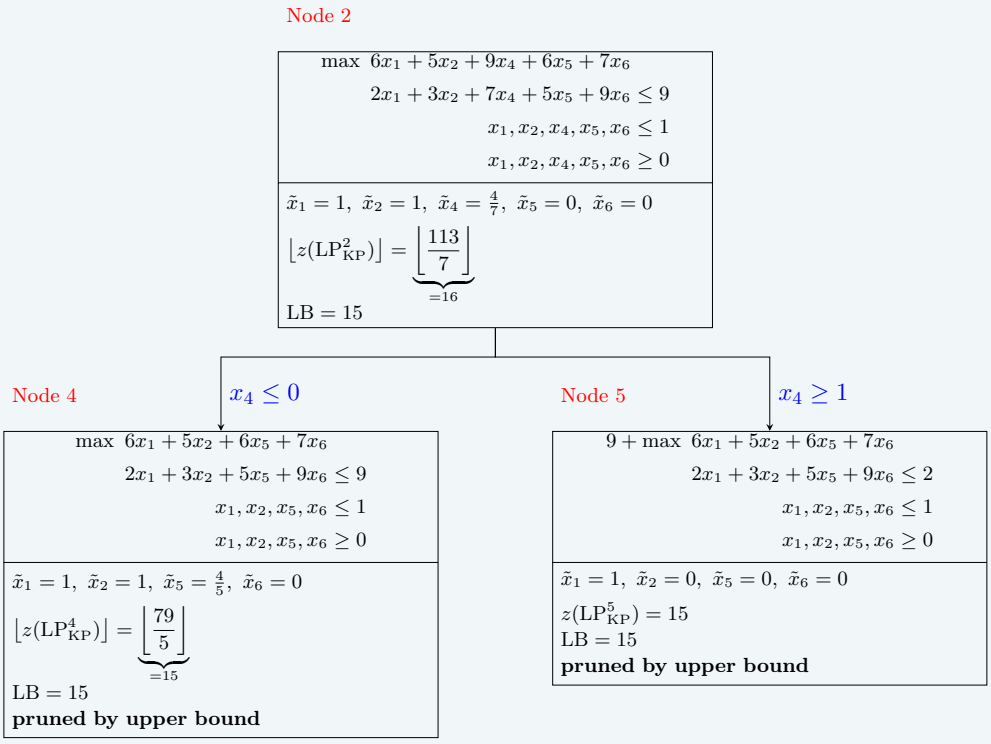
$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0 \quad \text{and} \quad z(\,\mathrm{LP}^5_{\mathrm{KP}}\,) = 9 + 6 \cdot 1 = 15$$

The $\mathrm{LP}^5_{\mathrm{KP}}$ problem is **feasible** and we have:

$$\lfloor z(\mathrm{LP}^5_{\mathrm{KP}}) \rfloor \leq \mathrm{LB} \qquad \text{since} \qquad 15 \leq 15$$

accordingly, the branching node is pruned by upper bound. The optimal solution of the $\mathrm{LP}^5_{\mathrm{KP}}$ problem is also integer, but its value does not improve the incumbent solution: the node is pruned by bound, as the other nodes of this example.

At the end of the second iteration, Node 4 and Node 5 are both pruned by upper bound. The corresponding branch-and-bound tree is:

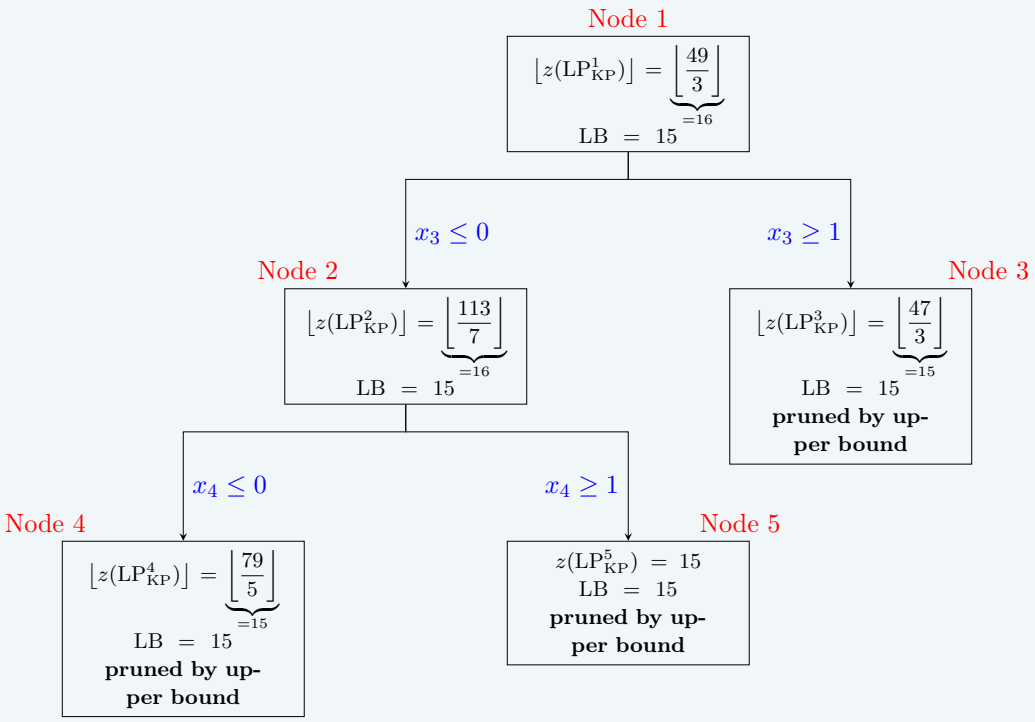


- **Termination**

  Since the problem list L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\text{ILP}_{\text{KP}}) = 15$ (the value of the lower bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\text{ILP}_{\text{KP}})$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1, \tilde{x}_5 = 0, \tilde{x}_6 = 0$ (the incumbent solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

This is the branch-and-bound tree of the previous B&B Algorithm, showing for each branching node the associated LP_{KP} problem, its optimal solution and value, the lower bound LB, and the pruning criteria.



29.3 Typical mistakes

Warning

- **Forgetting the upper bounds $x_j \leq 1$.** Without them the LP problem of a node would put all the capacity on the item of best efficiency: the node boxes must contain $x_j \leq 1$.
- **Computing the split item on the original instance.** In a node, the split item and the residual capacity are computed on the items of $\mathscr{I}^k$ with the capacity $c(k)$, after fixing the items of $\mathscr{I}^k_{\text{in}}$ and $\mathscr{I}^k_{\text{out}}$.
- **Forgetting the constant $p(k)$.** The value of the node is $p(k)$ plus the value of the items of $\mathscr{I}^k$: the profit of the selected items counts.
- **Using the old lower bound.** The lower bound is updated as soon as a node has an integer solution of larger value; the nodes of the list that become dominated are discarded.
- **Numbering the skipped children.** A child that is not solved does not receive a node index, exactly as in the general algorithm.

29.4 In summary

Summary

- The branch-and-bound algorithm for the KP is the LP-based algorithm with two simplifications: the problem of every node is solved by the formulas of Dantzig, and the branching variable is the one of the split item.
- A node is described by the sets $\mathscr{I}^k_{\text{in}}$, $\mathscr{I}^k_{\text{out}}$ and $\mathscr{I}^k$, by the capacity $c(k)$ and by the constant $p(k)$.
- The bound of a node is $\lfloor z(\text{LP}^k_{\text{KP}}) \rfloor$; when the problem list is empty the lower and the upper bound coincide and the incumbent solution is optimal.

Dashboard of the KP instance of Example 1 ($c = 9$, $p = (6, 5, 8, 9, 6, 7)$, $w = (2, 3, 6, 7, 5, 9)$). The bounds on $z(\text{ILP}_{\text{KP}})$ found so far:

Source	Lower bound	Upper bound	From
Dantzig bound	–	16	$\lfloor \frac{49}{3} \rfloor$
This chapter, initialization	15	16	heuristic solution $\{1, 4\}$
This chapter, iteration 1	15	16	node 2 in the list, $\lfloor \frac{113}{7} \rfloor$
This chapter, termination	15	15	empty list, 5 nodes